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by

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A dissertation submitted in partial fulfillment
of the requirements for the degree

of

Doctor of Philosophy

in

Engineering

MONTANA STATE UNIVERSITY
Bozeman, Montana

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NOMENCLATURE

μ Dynamic viscosity
$\mathbf{n}$ Normal vector
$\mathbf{u}$ Velocity vector

ABSTRACT

The abstract must be single spaced and no more than 350 words. The abstract must contain the following elements: (1) statement of the problem, (2) procedure or methods, (3) results, and (4) conclusions. Mathematical formulas, abbreviations, diagrams, and other illustrative materials should not be included. It should be written to be understood by a person who does not have expertise in the field.

1

INTRODUCTION

Modern manufacturing relies heavily on producing precise, intricate metal components. This is often achieved through subtractive manufacturing techniques using equipment such as milling machines and lathes. These Machines have been in use for approximately two hundred years[Groover and Moriwaki], and during this time have been continually improved. Manual control of these Machines has incrementally replaced by increasingly capable computerization [Reintjes, Groover]. Modern milling Machines and lathes often use Computer Numerical Control (CNC), which removes the need for a human machinist to control each detail of the machining process through manual inputs. Due to this continued computerization, these machines have grown in complexity, enabling more advanced machining operations, including advanced work with ductile materials, brittle materials, and increasing complex part geometry. [Civiello, Groover] Continued development has also introduced new systems to these Machines, including systems that allow the Machine to autonomously detect in real time a broken tool and replace the tool, detect abnormally high stress loads, or monitor motor loads across the machine. These innovations are significant because they enable the Machines to autonomously catch in real time errors that previously would have been caught by a human, and with CNC Machines could have propagated into damaged parts or damaged Machines. These innovations work to reduce scrap part rates, damaged tooling, and critically machine downtime.

Machining Overview

Lathe

A lathe performs subtractive manufacturing by rotating the workpiece around the spindle axis, designated as the Z axis, while translating the cutting tool in the X and Z axes. The X axis is commonly used to move the tool radially and controls the finished diameter of the workpiece. The Z axis moves along the length of the workpiece and is used to control the axial extent of the cut. Given the workpiece rotates while the tool traverses a plane containing the axis of rotation, all features are produced is a surface revolution and axially symmetrical about the Z axis. [Citation needed for the standard lathe coordinate system and machining structure] Lathes are commonly used to achieve part geometries that require turning outer diameters, boring internal diameters, facing, and drilling and tapping holes along the Z axis. Contoured profiles, including hemispherical and other curved forms are produced by coordinating the X and Z motions simultaneously. A conventional lathe has no Y axis, the Y axis is traditionally used only on mill-turn machines and multi-axis turning centers. The Y axis enables the off axis of rotation features which would break axial symmetry. [Citation needed for the Y axis in a lathe]. During turning passes, the tool remains engaged with the workpiece continuously, therefore unless in absence of material variation or progressive tool wear, the cutting force is approximately consistent. This force is conventionally resolved into three orthogonal components, tangential, radial, and feed. [citation needed for force directions] The magnitude of each of these forces is determined largely by the tool geometry, machine geometry, and the part characteristics. Large contributions to this are determined by the tool geometry and the lead and rake angles. [Citation needed for the load vectors and definitions]

Mill

A milling machine performs subtractive manufacturing by rotating the cutting tool about the spindle axis, designated the Z axis in a milling machine. Part geometry is achieved by moving the tool and workpiece relative to each other along three orthogonal directions. On a standard vertical machining center, the table carries the workpiece in the X and Y axes while the spindle head provides motion in the Z axis setting the depth of cut. Through these three degrees of freedom, material can be removed as arbitrary locations on the workpiece, so a far wider range of geometry can be produced compared to a lathe. Additional rotary axes extend this part geometry further. Four-axis machines add a single rotary axis about the X or Y axis and five-axis machines add a second rotary axis allowing the tool to approach the workpiece at a continuously variable orientation and make undercut and complex contoured surfaces without rotational symmetry possible. [Axis definitions and part geometry citation needed] The cutting force in a milling operation is applied to the workpiece through the flutes of the rotating tool, and each flute enters and leaves the cut once per spindle rotation. Engagement therefore is intermittent rather than continuous, and the load present on the machine is periodic at the tooth-passing frequency. The tooth-passing frequency is equal to the spindle rotational speed multiplied by the number of flutes on the cutting tool. [Citation needed for the rotational properties, endmill definitions, and intermittent load characteristics]

CNC Based Manufacturing

Both lathes and mills can be improved by the addition of computer based control, and machines of this type are known as Computer Numerical Control (CNC) lathes and mills. Through CNC, tool paths are specified as coordinated motion commands rather than performed by an operator adjusting hand-wheels and cranks, CNC toolpaths allow for simultaneous coordinated multi-axis motions that are required for complex contoured

geometry that would not be practical when performed by a human operator using hand-wheels and cranks to control the machine. [Citation needed regarding the additional complex geometries enabled by CNC] On servo based machines, motion is regulated by nested closed-loop position and velocity controllers acting in each axis [Citation needed regarding modern control mechanisms for servo based machines] Because the controller compares the commanded position against measured position at each update of the servo loop, often occurring faster than the machine could reach the desired location, position error is corrected for continuously and is therefore corrected. This control system allows CNC machines to hold a high level of precision and to reproduce the same result across many parts. Machines driven by stepper motors are commonly used and typically operated in an open-loop configuration. This principle takes advantage of the internal stepping and gearing motions a stepper motor provides to achieve reproducible and precise control at a lower level than servo based machines, but at a lower cost and reduced complexity. [Citation needed about different drive types, especially the difference in performance of stepper based control.] Adding sensing beyond the position control loop permits tighter control and the recording of process data during machining. This data can be retained on a per-part and per-machine basis and support process improvements across a production run in a manner that manual machining did not allow for. [Citation on data collection and process manufacturing improvements made to machines] High-performance CNC machines are typically expensive and occupy large floor footprints. These large often cast based footprints are used to raise the mass and structural dampening of the machine, attenuating vibration and disturbance forces before tool disturbance can occur. This contributes substantially to the precision and repeatability these machines can achieve [Citation regarding machine vibrations and precision needed] If these disturbances can instead by characterized and rejected within the control system, a portion of the precision obtained by this structural mass could in principle be obtained form control, allowing smaller and lighter machines to approach the performance

of larger cast based machines. Machines of this type would be usable in applications where the machine must be enclosed and where cost or space required for this enclosure currently constrains what can be produced.

Observability of Machining Loads

Some modern CNC machines are capable of recording substantial data during operation, and that data is presently used in three primary ways: to improve part quality across a production run, to reduce machine cycle time, and to indicate when maintenance should be performed. Each of these acts after the fact, as the part being cut when the data was collected is already complete. With additional sensing and characterization of the machine, the same information could support correction during the current cut, acting on the workpiece currently being machining rather than the next one. Corrections of this kind would also accumulate, producing a record of which in-process adjustments succeeded which could be applied to a new part geometry without repeating the same trial cuts. Systems performing monitoring of this kind are already commercially available. The Tool Monitoring Adaptive Control (TMAC), developed by Caron Engineering interfaces directly with a range of CNC controls, including Okuma platforms. The system monitors spindle motor power during the cut and operates on the principle that the power required to cut increases as the cutting edge wears. It applies this logic in two ways, it retires a worn tool or halts the machine on breakage, and in adaptive control mode, it overrides the programmed feed rate to hold spindle motor power near a set limit. [Cite Sam's work on TMAC] Data from these cuts can be retained for later analysis and process improvements. [Cite Sam's work again] The adaptive capability therefore exists, but it acts on an aggregate quantity. Spindle motor power represents the total power drawn by the cut. It does not resolve the force at the cutting edge into its principle components, and holding that power near a limit regulates the process only in the coarsest sense, without reference to the cutting conditions that have produced these changes.

[Cite Sam's thesis] Through the work completed in, [Sam's thesis source number] found that cutting force at the cutting edge could not be resolved from the data accurately enough to infer the resulting change in part geometry or surface finish. This limitation was attributed to the sample rate of the TMAC system, where the noise was no longer distinguishable from the signal. The signal was sampled at a rate of 200Hz which was well below the floor of the noise. This resulted in the data that showed the process was feasible, however the electrical sampling system would need significant changes to be practical. A system with direct access to the motor's electrical quantities and closer integration with the motor may resolve the signal to noise ration problem and provide more insight than the TMAC system could. With advances in modern CNC manufacturing equipment, new data can be collected. Currently this data can be used for process improvements to improve part quality over the life cycle of a part, decrease machine run time for part completion, or to infer when maintenance should be completed. With additional sensing and machine characterization, in situ changes could be made to prevent the current workpiece from failure rather than waiting for part completion to make process improvements on the large scale. This focus on improving the current part can be used to build up a smaller scale of improvements that happen live eventually building to a repertoire of changes that can be referenced on new part geometry.

Machine Load Transmission

When an axis is driven by a ballscrew, the cutting force acting on the carriage produces an axial load on the ballscrew. Given that a ballscrew is backdrivable the axial load appears at the motor as a rotational torque. The magnitude of this torque is set by the screw lead and the backdriving efficiency of the drive. [Machine design/Ballscrew dynamics citation needed] The cutting load therefore reaches the motor through the existing drive system, without additional instrumentation added to the machine. Many CNC machines come standard with ballscrews, however those that use alternative drive types such as lead

screws can be retrofitted with ballscrews to achieve this behavior and enable this torque transmission. Electric motors produce torque as a function of supplied current, the form of this current torque relationship varies on motor design. However, all electric motors fundamentally convert electrical current into a magnetic field that is used to produce torque. Some motor types produce torque at a fixed linear relationship to current, while other are inferred through a non-linear relationship dependent on rotor angle. Determining the load from the current therefore requires a priori knowledge on the electrical characteristics of the motor. This torque to current relationship motivates the review of motor types in the following section. Where the relationship is known, the cutting load appears at the axis drive as a disturbance torque, and the controller responds by changing the command current in order to preserve the axis trajectory. The cutting load is consequently present in the current signal and can be recovered from this disturbance. Recovery of this cutting load requires separating this load from the other torques the motor supplies during the same motion. These additional disturbance torques include friction in the system, inertial torque required to accelerate and decelerate the axis, and on a vertical axis the effects of gravity on the load of the carriage. Inertial and gravitational contributions follow from the axis mass and command trajectory can be predicted directly and therefore identified as a known disturbance. Friction is less traceable, comprising Coulomb, viscous, and velocity dependent component that vary with temperature and exhibit a hysteresis at velocity change. [Citation needed for friction characteristics and ballscrew motor dynamics] These disturbance therefore instead of being predicted are instead estimated and corrected for in situ through observer-based and signal processing techniques present in the proposed control system. [Citation needed for friction estimation with observers and the ADRC] The underlying principle for torque based disturbance characterization is already applied in a limited form on modern CNC machines. This is used where drive current detects events without any added sensor. These methods are commonly described as “sensorless”. The term originates in motor control,

where it denotes the estimation of rotor position without a position sensor. In this context is is used in the broader sense adopted in machine tool practice, meaning that the quantity of interest is inferred from signals already present in the drive rather than measured by a sensor added for that purpose. Homing against a hard stop is one example. When the axis reaches the stop, the motion ceases while the controller continues to the command position. This position error accumulates and the commanded current rises toward the motor limit. This sustained rise identifies a hard stop and can be used to establish a machine reference point without a limit switch. [Citation needed for this technology. Used in 3D printer and specifically BambuLabs and Voron printers] This hard stop referencing method is only practical due to the relatively high speeds at which the controller is able to sample error and detect a disturbance so no damage to the machine occurs during this brief period of time when the hard stop is in contact with the carriage on that axis. This principle can be expanded to detect other machine changes, tool breakage is another common use case for detecting sudden current load changes. This is normally performed by monitoring the spindle drive rather than an axis drive. The sudden loss of a cutting edge produces an abrupt decrease in spindle load, most commonly a decrease in power as the tool is no longer in contact with the workpiece. [Citation for tool breakage detection needed] This signal is drawn from a different motor, however the principle is the same in both use cases. Both the axial hard stop and the tool breakage are determined by sudden change in current draw in the motor controlling that motion. The present work extends this same principle from an event detection approach to a continuous estimation of the load magnitude.

Characterizing Load at the Cutting Edge

With a motor and control system of sufficient electrical and control bandwidth, current can be sampled rapidly enough to follow variation in the cutting load rather than only tracking an average value. Turning is the appropriate operation in which to establish this

relationship first. The tool remains engaged continuously through a turning pass, and during a hemispherical operation translates in both the X and Z axes causing a change in the force vector direction. With these passes, in the absence of material variation or progressive tool wear, the cutting force is approximately steady and the motor torque is likewise steady. Milling by contrast, presents an intermittent load at the tooth-passing frequency, superimposing a large period component on the signal of interest. Beginning with the turning isolates the current-to-force relationship from the periodic relationship which allows it to be established under the simplest available loading condition. Direct instrumentation of the cutting zone is not viable in the target applications, where the machining environment is enclosed and access is limited. Introducing sensors into this environment introduces a non-inconsequential cost aspect and would expose sensors to potential damage due to conditions internal to the machine during operation. Using already existing mechanics to inter the load from the motor current instead, locates the measurement at the drive, where the necessary electrical quantities are already present in the controller, and requires no instrumentation in the cutting zone at all. Establishing this relationship also opens a path to quantity that is not otherwise observable during machining. Force at the cutting edge is distributed over the contact between the tool and the workpiece, and the pressure in that contact, rather than the total force, is what governs the material's response to the cut. Pressure follows from force and the area over which it acts, and that area is determined by the chip cross-section and geometry of the cutting tool. [Citation for chip formation, and Sam's work on pressure to cutting force] Specific cutting energy provides the standard basis for relating the measured force to the condition in the cutting zone. [Cite Sam's thesis again] Where force can be recovered from drive current, tool geometry, and chip cross-section from machining parameters, an estimate of pressure at the cutting edge can be determined without additional measurement. A pressure estimate obtained through this could be of use where a material's machining envelope depends on the loading conditions it experiences

rather than on nominal properties alone. Certain material classes undergo solid-state phase transforms within the range of stress and temperature encountered at the cutting edge, and for these deformations the behavior is dependent on the pressure and temperature history the material has experienced. [Citation needed for manufacturing HE] Machining such materials predictably required a priori knowledge of the load actually being applied at the cutting edge, which no commercial machine presently provides. Recovering pressure in this manner is beyond the scope of the present work, which establishes a relationship between motor current and axis load that any such estimate would depend upon. This approach is identified here because it defines the direction in which this capability could extend, and because the requirements this imposes depend on the method of data collection in the present work. This includes, bandwidth sufficient to follow the cut, a torque-to-current relationship, and a control system capable of accepting current feedback. Meeting these requirements on a stepper-driven axis requires a motor model, a control architecture capable of maintaining the torque-to-current relationship under load, and a processing platform capable of operating at the required rate. Each of which is developed in the present work.

Electrical Overview

To accurately capture the present work, the fundamentals of common motor types are reviewed. Motor types determine the torque-to-current relationship. Direct Current (DC) motors have a relationship such that the torque produced is directly proportional to the current passing through the motor windings. The torque to current relationship in a DC motor does not depend on the motor rotor angle for torque generation and is constant regardless of the current rotor angle. Brushless Direct Current (BLDC) motors and Hybrid Stepper Motors do depend on the rotor angle as part of the current to torque output relationship along with the current moving through the motor windings.

DC Motor

DC Motors have the most direct relationship for current and torque. In a DC motor, a voltage is applied to the motor windings which are located on the rotor, and permanent magnets inside respond to the generated electric field and rotation is generated. While this motor type provides a simple installation and simple operation, the mechanical link of using brushes to pass the current from the fixed motor connections through the commutator to the rotating rotor is susceptible to failure from wear and can lead to a shorten motor life compared to technologies such as BLDC motors.

(Insert figure of the internals of a regular DC Motor)

Brushless DC Motors

Brushless DC Motors are often used in servo-based CNC machines where the longevity and precision control of the axes are required. In these applications a BLDC motor is selected due to the lack of brushes internal to the motor that can degrade with use. With no direct contact between the rotor and the stator, BLDC motors are not susceptible to the same wear and degradation as a DC motor with brushless in the commutator failing. Therefore BLDC motors have less wear internally that cause premature motor failure. These motors require additional control to operate as these motors require timing to three separate internal phases to properly generate rotational motion. BLDC motors differ from Permanent Magnet Synchronous Motors (PMSM) in the way that the motors are driven and the torque output relationship. In PMSM motors the coil windings are such that an AC signal is required to drive the motor, meaning external sinusoidal control of the voltage is needed. Where as with the BLDC motor, the motion can be generated with a trapezoidal method instead of a sinusoidal method.

(Insert figure of the internals of a brushless DC motor)

Hybrid Stepper Motors

Hybrid Stepper Motors are used when repeatable motion is desired at a low cost. The hybrid stepper motor takes advantage of internal fixed coils that cause predictable and repeatable rotation in a step motion. The internal rotor can be thought of as moving akin to a gear where each coil will advance the gear by one quarter of a tooth. While this is not the direct method used in hybrid stepper motors, this model provides a more clear understanding of the intended use case for the stepper motor. Internal to a stepper motor the rotor and stator poles are set such that both magnetically align at each step at the maximum magnetic field strength. This electrical and magnetic coupling is what provides the strong holding force of a stepper motor along with the stepping action. A standard method to driving a stepper motor is to use pulse trains alternative between the two phases of the motor. This electrical signal is simple to produce and causes rotational motion of a stepper motor in a way that can be easily discretized.

(Insert figure of Hybrid Stepper Motor)

Field Oriented Control

Field Oriented Control Field Oriented Control (FOC) is commonly used on BLDC motors and three-phase power systems to transform the multiple phases into two phases; the Quadrature and Direct phase, the Quadrature phase is the torque generating phase that is always leading the electrical angle by 90 degrees while the Direct phase is aligned with the rotor flux and therefore is not producing a rotational torque. This method is commonly used on BLDC motors as FOC reduces the complexity of driving the BLDC motor down to two vectors, the Quadrature which is scaled by the required torque output and the Direct which is minimized. Through this transform, the motors are able to maximize the torque produced and linearize the torque-to-current transform.

(Include figure of BLDC through Park transform)

The present work sets to show that this relationship can be extended from the BLDC motor to the Hybrid Stepper Motor. The same fundamental property is being used to convert the discrete steps internal to the stepper motor into a continuously rotating system. The Quadrature current is the current solely responsible for generating rotational torque in the stepper motor and the direct current is wasted energy that does not contribute to the rotation or stepping motion of the motor. This is because the Direct current generates a flux along the rotor magnet axis which does not generate torque but still passes current through the motor. Through this transform, the Hybrid Stepper Motor will be treated as a synchronous machine where two continuous sine waves can be applied to the two phases of the stepper motor at 90 degrees out of phase to generate rotation. The phase of both sine waves is directly linked to the motor rotor electrical angle keeping both in phase with the motor to generate the rotation.

(Include figure of Clarke Transform and Stepper Motor)

Control Systems Overview

The current work covers two control systems which necessitates a review the basics of both control systems. Traditional control systems are computed and designed in the frequency domain while modern control systems are often completed in the time domain through the use of linear algebra. The initial design for implementing FOC control on a stepper motor was achieved through a cascaded PID control loop. This provided some initial results and proved the system was worth pursuing further. One of the limitations of this solution came from the hardware running it, a RaspberryPi 4. This introduced inconsistencies in timing and created a non-deterministic control system. Due to these computational limits, the system was moved to an FPGA which also provided the possibility of doing computation using matrices while having deterministic timing. With this, the present work uses an Active Disturbance Rejection Control method to remove additional

disturbances in the system through the use of observers. Despite the advantage of having matrices to represent the control law of the system, a minimum footprint Active Disturbance Rejection Control algorithm was used. This converted the matrix mathematics to a transfer function that could be more efficiently simulated.

Proportional Integral Derivative Control

One classical approach to control systems for CNC Machines is to use Proportional Integral Derivative (PID) control to control the motion of the CNC machine. A PID system makes use of comparing the feedback to the target reference point to calculate the error. This traditional control system is often implemented as a single input single output system (SISO) where the position and velocity control are handled separately. [Add diagram of PID controller along with the equations that describe the PID controller] Through this a position motion profile can be created where the velocity control system is quickly saturated to the reference feedrate specified in the G-Code for the program. Through this, a responsive system that is able to reach the target position which a high level of accuracy is achieved while maintaining the desired feedrate on the machine. This modification of a traditional position control system allows for a set velocity target to be maintained during cutting operations which is vital to the proper functionality of a milling machine or lathe. Constant cutting velocity provides a consistent chip load which produces a more precise final result. [Add diagram of the cascaded PID system showing the position to velocity handoff] To achieve FOC control using PID systems, a preliminary design was created that utilized a PID controller for the position, velocity, quadrature current, and direct current. This system involved cascading the control system such that the output of one controller became the reference for the next controller. Through this, a reference quadrature current would be established that would provide the correct current to generate the target velocity and move the motor to the desired location at the desired velocity. By using the Park Transform,

the quadrature current was made linear in respect to the output torque. The quadrature current reference is treated as a step target with a constant current being related to a constant acceleration while under no-load operation. During a constant load, instead of the quadrature current settling at zero with no motion, the quadrature current will settle at a value directly proportional to the load on the system. Running in parallel to the quadrature current control loop was the direct current control loop with a constant reference input of zero. Through this, a linear current target change can be attributed to disturbances in the motion of the motor such as an increase of torque working against the rotation of the stepper motor. [Add diagram of the current control loop for PID] Early success was found with this system, however this was implemented on a RaspberryPi 4 B 8Gb model which introduced limitations in how fast the control system could update and how fast feedback could be collected. This limited the control system to updating at approximately 200Hz. (Need to add the math behind this at a later time) This limitation caused for the time constant of the stepper motor to be faster than the rate at which the control system could update. Through this, no meaningful change in current could be detected that could be related to the change in torque observed. However, this initial system did prove that a stepper motor could be run using FOC and that a change in quadrature current was noticeable in the offset of the waveform as the load increased. Given limitations in the software system on the RaspberryPi and potential additional disturbances in the mechanical system imperfections, a true relationship between the current and the torque could not be derived.

Active Disturbance Rejection Control

A principle limitation of PID control is the delay in detecting disturbances through the error signal from the plant. With PID control, there is no predictive work being done to estimate the error that will result from the plant and how to separate out error from the plant versus error that is the result of a disturbance. With the Active Disturbance Rejection

Control (ADRC) the error of the plant is predicted through an extended state observer (ESO). Modeling this error using the current reading from the plant allow for converging the estimate of the plant response to the control signal. Through this, the way the plant will react to a given control signal can be efficiently modeled and disturbances to this control signal predicted. By using an ADRC system, the disturbances of the plant can be modeled and removed which allows for a clean control signal to be provided to the plant. This results in a quadrature current reference that is more predictable and will account for external loads that were not modeled in the stepper motor. To achieve this result, a minimum footprint ADRC system was modeled in Simulink where an ADRC is used to control the velocity of the motor and the current in both the quadrature and direct phase. A proportional controller is used to control the position of the motor, as the output of the position controller following convention should saturate to the velocity limit imposed by the machining operation.

FPGA Overview

TODO: Need to add an overview of the FPGA and how it is being used.

FIELD ORIENTED CONTROL WITH TWO-PHASE HYBRID STEPPER MOTOR

Under FOC a hybrid stepper motor treated as a PMSM, the quadrature current in the motor is directly linear with the load torque with a slope of Kt. Through this relationship, the torque load currently experienced by the motor can be characterized and predicted. To test this relationship, a Simulink model was constructed using an ADRC based control system and a stepper motor model that matches a Nema 23 stepper motor. The original model was produced as part of a previous thesis focused on the analysis of the resource utilization of different control systems on an FPGA. The model that was produced for that work did not simulate with errors left in the fixed point precision, timing, and broken parameters inside the ADRC models. Additionally, this model was purely a mathematical representation of ADRC and stepper motor, there was no logic present for the handling of real-world signals that would be required to integrate the stepper motor with the FPGA. At this time, the model did not use PWM, ADC codes, and encoder pulses. The model relied on providing analog voltages to the stepper motor and receiving feedback in the form of radians and amps. This feedback is not directly accessible on the designed test bench and therefore had to be modified to be compatible with the PCB and mechanical test bench. From here, a PWM generation model, ADC decoding model, and encoder pulse handling logic was built to replicate the work done on the PCB. This included applying the same filters that are present in analog systems to the Simulink model, and modeling the sensors to provide real-world feedback signals. The goal of this work was to accurately model the complete system under test such that the data collected from the simulation would be reflective of real-world data.

System Architecture

The model is split into two primary parts, the control system and the stepper motor. The stepper motor model includes the stepper motor and the emulation of loadcell, ADCs,

and the rotary encoder. This half of the simulation is not designed to be synthesized into HDL, this is purely constructed for simulation purposes. The control system model is designed to be synthesized into HDL, this half of the simulation contains the ADRC controllers, and the logic to process the raw data from the ADCs and encoder. Additionally, the output of the controller supplies a PWM waveform that can be directly output from the FPGA to the PCB. This work moved forward with replicating the physical system as best as possible in the simulation such that the simulation would produce a near 1-to-1 replica of the physical system. The work was then expanded to include modeling of the ballscrew connected to the stepper motor to model how a spring would apply a linear load to the ballscrew creating a torque on the stepper motor. Once this was completed, spring loads were applied to the simulated system allowing for testing the relationship between the quadrature current and the torque of the motor.

The PWM is generated as a boolean value that is turned on and off based on a counter running at 200MHz, which provides a resolution of 0.012 Volts with a carrier frequency of 50KHz. The direction is set through a second boolean. These two outputs per phase allow for driving a standard half-bridge motor controller to control each phase of the stepper motor independently.

The stepper motor model provides as feedback, three 16-bit ADC codes, booleans for the rotary encoder A and B phase pulse and a boolean for the Z index of the rotary encoder. The current in each phase is reported on two of the three ADCs, and the third provides the voltage from the loadcell.

During debugging of the model, the stepper motor model was setup to provide the actual angular position of the rotor, angular velocity, and electrical angle. These values were used to confirm the simulation was properly processing the values through the emulated sensors and reconstruction. These values are unused in the control system.

(Figure of the system at a high level (More than a flow chart but not Simulink))

Prior Work and Contributions

The original continuous time, double floating point model was created by Dr. Edens, which was then revised by Caleb Binfet to include solving in the discrete time and partial work on the fixed point conversion. From here, Caleb shows in [Caleb's Thesis Reference] that the first order MFP ADRC used XX amount of resources on the AMD KR260 Robotics Starter Kit SoM FPGA. Through this decrease in resource utilization it was an ideal choice to provide the disturbance rejection for parameters of the stepper motor and testing that could not accurately be captured in a simulation. Such as, friction, bearing imperfections, linear rail imperfections, ball screw imperfections, copper voltage loss. When this model was transferred to begin work on integration with the PCB and mechanical system, the provided model did not simulate. This lead to the work done to fix the model and adapt the model to work with the the PCB. During this, I rebuilt the discrete time system to fix integration issues, reworked the fixed point precision to use standardized values avoiding conflicts in future work, and proceeded to build out the additional peripheral emulations. This system represents the most complete model of the stepper motor created in the SE3 lab and provided a fourth iteration on showing the relationship between the quadrature current and the torque produced. Initial work was based on an Arduino system that provided an initial proof of concept that the stepper motor would rotate with voltage as directed by the FOC algorithm. From here premade 3-phase BLDC FOC compatible motor drivers were tested to seek an off-the-shelf solution to the problem. Ultimately, these boards were not compatible with the stepper motor in the desired configuration and a custom implementation was created on a RaspberryPi. The RaspberryPi system proved that the correlation could be found and the entire algorithm would work end-to-end, however the clock speed of the RaspberryPi provided non-deterministic timing and led to difficulties in recreating data between experiments. From this, the FPGA based approach was determined and implementation began. This required reworking the original Simulink approach that

provided a cascaded PID system with continuous time and double floating point precision to a discrete time fixed point precision model. This provided an opportune time to implement the ADRC control approach instead of the PID approach.

Hybrid Stepper Motor Specifications

(Insert table with values from the Ding's Motion Datasheet) This specific Nema 23 class motor was selected due to the electrical characteristics being similar to the mini-tech CNC Mill in the SE3 lab. This mill has been used for precision machining work within the SE3 lab previously and provides a clear baseline to compare future work related to this system implemented on a mill for performance improvements.

Testbed

To test the performance of the stepper motor with FOC control, a mechanical test bench was constructed that provides dimensional constraints that match the X-Axis of the mini-tech CNC mill. This was selected to demonstrate the FOC algorithm would work over a full length of displacement for a mill compared to the six inches of displacement present on the RaspberryPi based test bench. This new test bench provides a system designed to isolate some of the friction issues present on the previous system and provide a repeatable method to test different load conditions of the motor. The mechanical test bench features a XYZ ballscrew, XYZ caged nut, and two XYZ load cells mounted in plane with the ballscrew to minimize the rotation caused by a spring on a single side of the ballscrew. The ballscrew has a 5mm lead, approximately 90% efficiency, with an inertia of $3.47 \times 10^{-5} kg \times m^2$ with the caged nut adding an inertia of $3.38 \times 10^{-7} kg \times m^2$. These parameters were provided by manufacturer of these parts, THK. These parameters were also modeled in the simulation. The two loadcells with XYZ parameters were used to capture the total load. In the simulation these two load cells are modeled as a single loadcell that fully captured the load.

(Figure: CAD Model of the mechanical test bench, label parts)

Through the use of these loadcells, the testbench can capture XYZ% of the total torque limit of the stepper motor. 2.3 Controller sensing and hardware Selected hardware for this work is the AMD KR260 Robotics Starter Kit SoM FPGA which was selected to be inline with work at Montana State University to move undergraduate courses to the AMD KR260 Robotics Starter Kit SoM FPGA.

→ More thought needed on exactly what to include about the PCB → More thought needed to determine what to say about the encoder and load cell selection reasons. → Expand on the rotary encoder with information from the datasheet → Create a table of all the sensors and important information such as resolution, range, quantity, etc.

Machining Application Constraints

With the future goal of applying this work to CNC machining applications, this imposed the goal of maintaining parameters that were indicative of real world machining loads. As such, the velocity was limited to 35.5 rads/second which on this system corresponds to approximately 339 RPM which is 66 inches per minute. On a machining application this would be a high feedrate to machine 6061 aluminum. Through this velocity limitation, the relationship between torque and current can still be determined which provides a start on future work that can implement the use of FOC control in machining applications. With the constrained choice of a nema 23 stepper motor to match the existing mini-tech mill, the current limit and voltage limit were imposed of 3A rms maximum continuous load and 48V limit. These constraints were modeled in the simulation to capture an accurate system.

Motor Parameters and Identification

Parameter Sources

The Dings' Motion Nema 23 Stepper Motor 23H2065 was selected for testing due to the electrical parameters of this motor being very similar to the Mini-Tech CNC Mill. The

reported electrical characteristics of both motors are listed in Table 1

Table 1: Electrical and mechanical parameters of the Dings Motion NEMA 23 stepper motor, model 23H2065.

Parameter	Symbol	Value	Tolerance / Condition
Rated current (per phase)	I_{rated}	3.0 A	RMS
Phase resistance	R	0.9 Ω	±10% at 20 °C
Phase inductance	L	2.7 mH	±20% at 1 kHz
Holding torque	T_{h}	1.60 N m	226.6 oz · in
Rotor inertia	J_{r}	350 g cm^2	—
Step angle	θ_{step}	1.8°	±5%
Motor length	L_{motor}	65 mm	Dimension "L"
Mass	m	860 g	—

Source: Dings Motion USA, NEMA 23 (size 23 / 57 mm) rotary stepper motor specification table.

Torque constant measurements

→ Setup for measuring Kt

Datasheet value for Kt Through the values provided above in the stepper motor datasheet, the expected Kt value can be calculated through: [Equation for Kt calculation]

This equation returns XYZ which is off from the value determined form the simulation by XYZ% and by XYZ% from the measured value of the stepper motor.

Model Parameter Integration

Modeling the plant was achieved through use of the Dings Motion 23H2065 datasheet, (Table 1) and the THK ballscrew catalog (model BIF1605-5RRGO+710LC5A) with a 15 mm shaft diameter and a 5 mm lead. The load-side inertia is reflected to the motor shaft as a rotational inertia. The additional inertia for the milling surface, workholding, and stock material was not modeled in this plant. The inertia change between a physical mill and the representative Simulink model will result in more inertia on a future system. This is not

expected to have substantial impact on the performance of the control system. A test was conducted by increasing the inertia of the plant in Simulink and comparing the results.

Torque Constant The torque constant for the plant was found through the datasheet and experimentally by *blank* method. To calculate the torque constant from the datasheet, the holding torque and the rated current was used in Equation ((2)). The provided current limit in the Dings Motion 23H2065 Datasheet is an RMS value which is converted into a peak value in Equation ((1))

$$I_{\mathrm{pk}} = \sqrt{2}\, I_{\mathrm{rated}} = \sqrt{2}\,(3.0\,\mathrm{A}) = 4.243\,\mathrm{A}, \tag{1}$$

$$K_t = \frac{T_{\mathrm{h}}}{\sqrt{2}\, I_{\mathrm{rated}}} = \frac{1.60\,\mathrm{N\,m}}{\sqrt{2}\,(3.0\,\mathrm{A})} = 0.377\,\mathrm{N\,m\,A^{-1}}. \tag{2}$$

Winding Resistance and Inductance The winding resistance and inductance are the per-phase datasheet values:

$$R = 0.9\,\Omega, \qquad L = 2.7\,\mathrm{mH}. \tag{3}$$

Therefore the electrical time constant is

$$\tau_e = \frac{L}{R} = \frac{2.7\,\mathrm{mH}}{0.9\,\Omega} = 3.0\,\mathrm{ms}. \tag{4}$$

Simulation work uses the provided datasheet values of 2.7 mH in calculation. The

inductance of the actual hybrid stepper motor was measured through a calculation of the impedance driven by a 1 kHz signal in series with a 15 Ω resistor. The values measured we within the expected expected tolerance range of 20% as listed in the datasheet. The values measured are:

Table 2: Measured stepper motor inductance

Phase	Symbol	Inductance
Phase A	L_A	3.205 mH
Phase B	L_B	3.279 mH

Source: Experimentally gathered, compared against Dings Motion USA, NEMA 23 (Size 23 / 57 mm) rotary stepper motor specifications table.

Rotor Inertia The motor inertia J_m is the rotor inertia provided on the datasheet (Table (1)):

$$J_m = 350\,\mathrm{g\,cm^2} = 350 \times 10^{-7}\,\mathrm{kg\,m^2} = 3.5 \times 10^{-5}\,\mathrm{kg\,m^2}. \tag{5}$$

Viscous Damping The viscous damping coefficient B_m is not provided on the datasheet for the 23H2065, for the purpose of simulation the viscous damping coefficient was estimated at a value that is less than will be expected on the physical system. This results in a simulation that is inherently more difficult to control than the physical plant will be.

$$B_m = 8 \times 10^{-4}\,\mathrm{N\,m\,s\,rad^{-1}}. \tag{6}$$

Ball Screw and Nut Inertia The screw shaft is modeled as a solid cylinder of outer diameter d and length L_s. With the catalog shaft mass per unit length $\lambda_s = 1.65\,\mathrm{kg\,m^{-1}}$ for

the $d = 20\,\text{mm}$ shaft, the shaft mass is $m_s = \lambda_s L_s$ and

$$J_s = \frac{1}{2}\, m_s \left(\frac{d}{2}\right)^2 = \frac{1}{2}\, \lambda_s L_s \left(\frac{d}{2}\right)^2. \tag{7}$$

For $L_s = 0.71\,\text{m}$,

$$J_s = \frac{1}{2}\,(1.65\,\text{kg}\,\text{m}^{-1})(0.71\,\text{m})(0.010\,\text{m})^2 = 5.8575 \times 10^{-5}\,\text{kg}\,\text{m}^2. \tag{8}$$

The nut translates as the screw turns. Its linear mass m_n is reflected to the screw shaft through the lead P_h, so that the kinetic energy of the nut equals that of an equivalent rotating inertia:

$$J_n = m_n \left(\frac{P_h}{2\pi}\right)^2 = (0.57\,\text{kg}) \left(\frac{0.005\,\text{m}}{2\pi}\right)^2 = 3.61 \times 10^{-7}\,\text{kg}\,\text{m}^2. \tag{9}$$

Total Inertia The total inertia seen by the motor is the sum of the three components:

$$J = J_m + J_s + J_n = 3.5 \times 10^{-5} + 3.47 \times 10^{-5} + 3.6 \times 10^{-7} \approx 7.0 \times 10^{-5}\,\text{kg}\,\text{m}^2. \tag{10}$$

Stepper Motor Model

To properly represent the control system and the expected functionality of the control system on the FPGA with the plant and PCB a detailed model of the plant and PCB were created. These models provided an accurate representation of the expected system combined

Table 3: Plant parameters used in the simulation and source of parameter.

Parameter	Symbol	Source	Value
Torque constant	K_t	$T_{\mathrm{h}}/(\sqrt{2}\, I_{\mathrm{rated}})$	$0.377\,\mathrm{N\,m\,A^{-1}}$
Resistance	R	Dings datasheet	$0.9\,\Omega$
Inductance	L	Dings datasheet	$2.7\,\mathrm{mH}$
Motor inertia	J_m	Dings datasheet	$3.5 \times 10^{-5}\,\mathrm{kg\,m^2}$
Viscous damping	B_m	Estimated	$8 \times 10^{-4}\,\mathrm{N\,m\,s\,rad^{-1}}$
Screw inertia	J_s	THK catalog, Eq. (7)	$3.47 \times 10^{-5}\,\mathrm{kg\,m^2}$
Nut inertia	J_n	THK catalog, Eq. (9)	$3.61 \times 10^{-7}\,\mathrm{kg\,m^2}$

with the expected data types and analog circuits.

Electrical Model

The electrical model of the stepper motor converts the logic-level pulse-width modulation (PWM) drive signals produced by the controller into phase voltages, and the phase voltages into phase currents. These phase currents in turn determine the electromagnetic torque acting on the rotor. The Simulink model is split into two levels, the logic circuits that emulate an H-Bridge motor driver and the electrical mechanics of the stepper motor. Figure shows the logic circuits diagram for the half-bridge motor driver, and Figure shows the subsystem that converts the motor drive voltage into phase currents.

Drive Stage Voltage Each phase of the stepper motor is driven independently with an H-Bridge to allow for control of the quadrature and direct vectors through a combination of each motor phase. For each phase $x \in \{a, b\}$, the controller supplies two Boolean signals, a PWM gate signal $u_x[k] \in \{0, 1\}$, which indicates whether the bridge is conducting at sample k, and a direction signal $d_x[k] \in \{0, 1\}$, which selects the polarity of the applied voltage. Table 4 summarizes these signals. The voltage synthesis stage maps each pair of signals to a phase voltage through three operations.

- **Type conversion.** A Data Type Conversion block casts u_x from Boolean to double

precision, so that all subsequent arithmetic is performed in floating point. Under this cast, a logical true maps to 1 and a logical false maps to 0. This is done as the stepper motor is not intended for HDL synthesis and is therefore modeled completely in double floating point.

- **Voltage scaling.** The control signal is multiplied by the motor source voltage to model the H-Bridge connecting the phase to the motor voltage. This is achieved through a gain block which assumes an ideal H-Bridge with no bouncing, ringing, or latency.

- **Polarity selection.** A Switch block with the criterion $d_x > 0$ passes the constant -1 when the direction signal is set and the constant $+1$ otherwise. This reflects the motor drive direction bit logic used on the physical H-Bridge drivers used to control the stepper motor.

Combining these operations, the voltage applied to phase x at sample k is

$$v_x[k] = K_v \, u_x[k] \, \sigma_x[k], \qquad \sigma_x[k] = \begin{cases} -1, & d_x[k] = 1, \\ +1, & d_x[k] = 0, \end{cases} \tag{11}$$

which can be represented compactly as $\sigma_x = 1 - 2d_x$. The phase voltage is able to take one of three levels, $v_x \in \{-K_v, 0, +K_v\}$.

When the gate signal u_x is low, Equation (11) applies zero volts across the winding regardless of the direction signal. This corresponds to the slow-decay mode of the physical H-Bridge, in which both low-side switches conduct and short the winding terminals together. With no applied voltage, the winding current evolves under the winding resistance and back EMF alone, as described by Equations (15) and (16) with $v_x = 0$, which produces a gradual braking torque on the rotor. The fast-decay mode, in which the winding current recirculates through the freewheeling diodes back into the supply, is not modeled. The simulated drive

therefore always operates in slow decay during the off portion of each PWM period.

Dead time, switch on-resistance, and freewheeling diode drops are omitted from this model, as the model assumes these disturbances will be handled by the Active Disturbance Rejection Control system controlling the quadrature and direct currents during run time.

The simulation sample time, T_s is restricted to be much shorter than the PWM period T_{pwm} so that the model resolves individual PWM pulses. Over one PWM period, 20 µs, with duty ratio D_x and constant direction, the average phase voltage is modeled as

$$\bar{v}_x = \frac{1}{T_{\text{pwm}}} \int_t^{t+T_{\text{pwm}}} v_x(\tau)\,\mathrm{d}\tau = K_v\, D_x\, \sigma_x. \tag{12}$$

Because the electrical time constant of the winding, shown in Equation (4), is normally much longer than T_{pwm}, the winding behaves as a low-pass filter. The phase current consequently follows $\bar{v}_x$, with a superimposed ripple. Neglecting resistance and back electromotive force (EMF), the peak-to-peak ripple is approximately

$$\Delta i_x \approx \frac{K_v\, D_x(1 - D_x)\, T_{\text{pwm}}}{L}, \tag{13}$$

which is largest at $D_x = 0.5$. This expression provides a check on whether the chosen PWM carrier frequency yields an acceptable current ripple for a given winding inductance L. A PWM carrier frequency that is not sufficiently fast would introduce discrete current behavior into the stepper motor, leading to discrepancies in the measured versus expected current values in the quadrature and direct phase.

Substituting the chosen simulation values into Equation (13) the peak-to-peak ripple is calculated to be approximately:

$$\Delta i_x \approx \frac{48 \times 0.5 \times (1 - 0.5) \times \left(\frac{1}{50000}\right)}{2.7 \times 10^{-3}}$$
$$\Delta i_x \approx 89\,\mathrm{mA} \tag{14}$$

A summary of the relevant electrical input signals is provided in Table 4.

Table 4: Input signals of the voltage synthesis stage.

Port	Signal	Type	Description
1	u_a (`Va`)	Boolean	PWM gate signal, phase A
2	u_b (`Vb`)	Boolean	PWM gate signal, phase B
3	d_a (`Va_direction`)	Boolean	Polarity select, phase A
4	d_b (`Vb_direction`)	Boolean	Polarity select, phase B

Phase Winding Model Each phase winding is modeled as a series resistance and inductance in series with a speed-dependent back EMF, which is the standard representation of a two-phase permanent-magnet or hybrid stepper motor [**? ?**]. The `StepperElectrical` subsystem receives the phase voltages v_a and v_b, the electrical rotor speed ω_r, and the electrical rotor angle θ_r, and it implements

$$L\frac{\mathrm{d}i_a}{\mathrm{d}t} = v_a - R\,i_a + K_\omega\,\omega_r \sin\theta_r, \tag{15}$$
$$L\frac{\mathrm{d}i_b}{\mathrm{d}t} = v_b - R\,i_b - K_\omega\,\omega_r \cos\theta_r, \tag{16}$$

where i_a and i_b are the phase currents, R is the winding resistance, L is the winding self-inductance, and K_Ω is the back-EMF gain, equal to the flux linkage ψ_m of Equation (26). The speed and angle inputs are the electrical quantities $\omega_r = \omega_e$ and $\theta_r = \theta_e$ of Equation (18).

Defining the back-EMF voltages as

$$e_a = -K_\Omega\,\omega_r \sin\theta_r, \qquad e_b = K_\Omega\,\omega_r \cos\theta_r, \tag{17}$$

both equations take the circuit form $v_x = R\,i_x + L\,\mathrm{d}i_x/\mathrm{d}t + e_x$. The sine and cosine dependence places the two back-EMF waveforms in quadrature, consistent with the 90° electrical displacement between the phase windings.

Electromechanical Model

Although the simulation is implemented in the stationary phase frame of Equations (15) and (16), the controller regulates the direct and quadrature currents, so the stepper motor is also described as a two-phase permanent-magnet machine in the rotor-fixed d–q frame. The motor has $N_r = 50$ rotor teeth, so one mechanical revolution corresponds to N_r electrical periods, and the electrical angle and speed are

$$\theta_e = N_r\,\theta_m, \qquad \omega_e = N_r\,\omega_m, \tag{18}$$

where θ_m and ω_m are the mechanical angle and angular velocity of the rotor. The phase currents are mapped into the rotor frame by the Park transform, which for a two-phase machine is a pure rotation by the electrical angle,

$$i_d = i_a \cos\theta_e + i_b \sin\theta_e, \qquad i_q = -i_a \sin\theta_e + i_b \cos\theta_e, \tag{19}$$

and the phase voltages and back-EMF voltages transform in the same way. With equal direct and quadrature inductances ($L_d = L_q = L$), applying Equation (19) to Equations (15)

and (16) gives the stator voltage equations

$$v_d = R\, i_d + L\, \frac{di_d}{dt} - N_r\, \omega_m\, L\, i_q, \tag{20}$$

$$v_q = R\, i_q + L\, \frac{di_q}{dt} + N_r\, \omega_m\, L\, i_d + N_r\, \omega_m\, \psi_m, \tag{21}$$

and the rotor dynamics are

$$J\, \frac{d\omega_m}{dt} = T_e - B_m\, \omega_m - T_L, \tag{22}$$

where T_e is the electromagnetic torque, B_m is the viscous damping coefficient, T_L is the load torque, and $J = J_m + J_s + J_n$ is the total inertia from Equation (10).

Phase Current Integration Within the `StepperElectrical` subsystem, Equations (15) and (16) are integrated with Discrete-Time Integrator blocks configured with the forward Euler method, the same method used for the rotor speed and angle in the mechanical model. For each phase, the subsystem forms the voltage across the winding inductance, $v_x - R\, i_x - e_x$, scales it by $1/L$ to obtain the rate of change of current, and integrates the result to obtain the phase current. Combining these operations, the phase current at sample $k+1$ is

$$i_x[k+1] = i_x[k] + \frac{T_s}{L}\Big(v_x[k] - R\, i_x[k] - e_x[k]\Big), \qquad x \in \{a, b\}, \tag{23}$$

where $v_x[k]$ is the synthesized phase voltage of Equation (11) and $e_x[k]$ is the back-EMF voltage of Equation (17), evaluated with the electrical rotor speed and angle at sample k. Collecting terms in i_x gives

$$i_x[k+1] = \left(1 - \frac{T_s}{\tau_e}\right) i_x[k] + \frac{T_s}{L}\Big(v_x[k] - e_x[k]\Big), \qquad \tau_e = \frac{L}{R}, \tag{24}$$

where τ_e is the electrical time constant of the winding from Equation (4). The single pole at $z = 1 - T_s/\tau_e$ is the forward Euler image of the continuous-time winding pole at $s = -1/\tau_e$, so the discretization is stable for $T_s < 2\tau_e$. The electrical time constant is typically far shorter than the mechanical time constant τ_m, so this condition, rather than the mechanical stability condition, sets the upper bound on the sample time. In practice the sample time is set by the stricter requirement that $T_s \ll T_{\text{pwm}}$ in order to resolve individual PWM pulses. Since $T_{\text{pwm}} \ll \tau_e$, the ordering $T_s \ll T_{\text{pwm}} \ll \tau_e$ holds, and the current integration is both stable and accurate with a wide margin.

The forward Euler method is also well suited to the PWM drive. During a single PWM pulse the applied voltage is constant, and when resistance and back EMF are neglected, as in the derivation of Equation (13), the current changes at a constant rate. Forward Euler integrates a constant rate exactly, so the simulated phase current reproduces the triangular ripple of Equation (13), with the timing of each pulse edge resolved to the nearest sample.

As in the mechanical model, each forward Euler integrator has no direct feedthrough, so $i_x[k+1]$ depends only on signals at sample k. The back-EMF term uses the rotor speed and angle at sample k, which are themselves integrator outputs of the mechanical model, and the electromagnetic torque at sample k uses the phase currents at sample k. No signal in the electrical–mechanical loop therefore depends on another signal at the same sample, and no algebraic loop is formed. The complete motor model is an explicit discrete-time update of the state vector $[\, i_a,\ i_b,\ \omega_m,\ \theta_m \,]^{\mathsf{T}}$, advanced once per sample by Equations (23), (31), and (33).

Flux Linkage and Torque Equation The flux linkage ψ_m is the peak flux that the rotor magnets link with one stator phase. It sets the back-EMF, the term $N_r\, \omega_m\, \psi_m$ in Eq. (21), which opposes the applied voltage and grows linearly with speed. Its units are V s.

The electromechanical power is the product of the back-EMF and the torque-producing current. Because the two-phase Park transform of Equation (19) is a pure rotation, it is both amplitude- and power-invariant, and the power into the back-EMF is $P_e = (N_r\,\omega_m\,\psi_m)\,i_q$, and since $P_e = T_e\,\omega_m$, the torque is

$$T_e = N_r\,\psi_m\,i_q = K_m\,i_q, \tag{25}$$

which defines the motor torque constant

$$K_m = N_r\,\psi_m \quad \Longleftrightarrow \quad \psi_m = \frac{K_m}{N_r}. \tag{26}$$

Here K_m is the same quantity as K_t in Eq. (2). Reluctance torque is absent because $L_d = L_q$, and detent torque is neglected. The direct-axis current i_d produces no torque and is regulated to zero.

For the 23H2065 motor, with $K_m = 0.377\,\mathrm{N\,m\,A^{-1}}$ and $N_r = 50$,

$$\psi_m = \frac{0.377\,\mathrm{N\,m\,A^{-1}}}{50} = 7.54 \times 10^{-3}\,\mathrm{V\,s}. \tag{27}$$

The factor of $\sqrt{2}$ in Eq. (2) arises because the datasheet holding torque is specified with both phases carrying the rated current, so the current vector has magnitude $\sqrt{2}\,I_{\mathrm{rated}}$. This vector corresponds to i_q when $i_d = 0$.

Mechanical Model

The mechanical model of the stepper motor converts the phase currents produced by the electrical model into electromagnetic torque and integrates the resulting torque balance to obtain the rotor speed and angle. These mechanical states are returned to the electrical

Table 5: Parameters of the electrical model.

Symbol	Simulink Block	Description
K_v	Voltage scaling gain	Motor supply voltage, 48 V
T_{pwm}	Phase A and B PWM Generators	PWM period, 20 µs (50 kHz)
R	`StepperElectrical`	Winding resistance (Ω)
L	`StepperElectrical`	Winding inductance, 2.7 mH
K_Ω	`StepperElectrical`	Back-EMF gain, equal to ψ_m, 7.54×10^{-3} V s

model, where they set the magnitude and phase of the back EMF, and are exported from the motor model for use by the encoder emulation. The Simulink implementation is divided into three subsystems, shown in Figure: the `Te` subsystem, which computes the electromagnetic torque from the phase currents and rotor angle; the `StepperMechanical` subsystem, which implements the rotor dynamics of Equation (22); and the `rotor angle` subsystem, which integrates the rotor speed to obtain the mechanical and electrical rotor angles. As with the electrical model, all signals are represented in double-precision floating point, since the motor model is not intended for HDL synthesis.

Electromagnetic Torque The `Te` subsystem receives the phase currents i_a and i_b from the electrical subsystem, `StepperElectrical`, and the mechanical rotor angle θ_m from the `rotor angle` subsystem. Because the electrical model is formulated in the stationary phase frame rather than the rotor-fixed d–q frame, the torque is computed directly from the phase A and phase B currents as

$$T_e = K_m \left[-i_a \sin\left(\theta_e\right) + i_b \cos\left(\theta_e\right) \right], \tag{28}$$

where the electrical angle $\theta_e = N_r \theta_m$ is formed within the subsystem from the supplied mechanical angle.

Equation (28) follows from the power balance of the back-EMF sources. Using the back-EMF voltages of Equation (17), the electrical power converted by the two phase windings is

$$P_e = e_a i_a + e_b i_b = K_\Omega \omega_r \left(-i_a \sin\theta_r + i_b \cos\theta_r\right). \tag{29}$$

Substituting the electrical speed $\omega_r = N_r \omega_m$ from Equation (18) and equating P_e to the mechanical power $T_e \omega_m$ gives

$$T_e = N_r K_\Omega \left(-i_a \sin\theta_r + i_b \cos\theta_r\right). \tag{30}$$

Identifying the back-EMF gain with the permanent-magnet flux linkage, $K_\Omega = \psi_m$, the leading factor becomes $N_r \psi_m = K_m$ by Equation (26), which recovers Equation (28).

The bracketed term in Equation (28) is the quadrature current i_q of the Park transform in Equation (19), so the phase-frame expression reduces to $T_e = K_m i_q$, consistent with Equation (25). The two formulations are therefore equivalent. The phase-frame form is used in simulation so that the torque responds to the instantaneous phase currents, including the PWM current ripple estimated in Equation (14), rather than to idealized d–q currents. As a result, the simulated torque carries a ripple on the order of $K_m \Delta i_x \approx 0.034\,\mathrm{N\,m}$ at the PWM carrier frequency, which the controller must tolerate. Detent torque is neglected, consistent with the electromechanical model.

Rotor Dynamics The `StepperMechanical` subsystem, shown in Figure, implements the rotor torque balance of Equation (22). The subsystem receives the electromagnetic torque T_e and the load torque T_L and produces the mechanical rotor speed ω_m through four operations.

- **Torque summation.** A Sum block with signs $(+,-,-)$ forms the net accelerating torque $T_e - B_m\omega_m - T_L$. The electromagnetic torque drives the rotor, while the viscous damping torque and the load torque oppose it.

- **Inertial scaling.** A gain block of value $1/J$ converts the net torque into angular acceleration, where J is the total inertia from Equation (10).

- **Integration.** A Discrete-Time Integrator configured with the forward Euler method, with transfer function $KT_s/(z-1)$ and unity gain K, integrates the angular acceleration to obtain the rotor speed.

- **Damping feedback.** The rotor speed is multiplied by the viscous damping coefficient B_m and returned to the negative input of the Sum block, closing the damping loop.

Combining these operations, the rotor speed at sample $k+1$ is

$$\omega_m[k+1] = \omega_m[k] + \frac{T_s}{J}\Big(T_e[k] - B_m\omega_m[k] - T_L[k]\Big), \tag{31}$$

which is the forward Euler discretization of Equation (22). Collecting terms in ω_m yields the discrete transfer function from net driving torque to rotor speed,

$$\frac{\omega_m(z)}{T_e(z) - T_L(z)} = \frac{T_s/J}{z - (1 - T_s/\tau_m)}, \qquad \tau_m = \frac{J}{B_m}, \tag{32}$$

where τ_m is the mechanical time constant of the rotor. This is the forward Euler image of the continuous-time plant $1/(Js + B_m)$, whose pole at $s = -1/\tau_m$ is mapped to $z = 1 - T_s/\tau_m$. The discretization is stable for $|1 - T_s/\tau_m| < 1$, which requires $T_s < 2\tau_m$. Because the sample time is chosen much shorter than the PWM period in order to resolve individual pulses, and the mechanical time constant is much longer than the PWM period, this condition is satisfied with a wide margin and the discretization error of the forward Euler method is negligible.

The forward Euler method was selected because it has no direct feedthrough: the integrator output at sample k depends only on inputs up to sample $k-1$. This prevents the damping feedback path through B_m from forming an algebraic loop with the Sum block. The same one-sample delay decouples the electrical and mechanical models, since a change in T_e at sample k affects ω_m only at sample $k+1$. Given that $T_s \ll \tau_m$, this delay has no meaningful effect on the simulated dynamics.

The load torque T_L is supplied externally through input port 3 of the motor model, which allows step or time-varying load disturbances to be applied during simulation. Because T_L enters the torque balance directly and is not measured by the controller, it appears to the Active Disturbance Rejection Control system as part of the total disturbance to be estimated and rejected. The rotor speed ω_m is routed to output port 4 of the motor model, to the `rotor angle` subsystem, and through a gain of N_r to form the electrical speed ω_r supplied to the `StepperElectrical` subsystem.

Rotor Angle The `rotor angle` subsystem, shown in Figure, integrates the rotor speed to obtain the rotor position. A second forward Euler Discrete-Time Integrator produces the mechanical angle, and a gain of N_r, labeled `RT` in the model, converts it to the electrical angle:

$$\theta_m[k+1] = \theta_m[k] + T_s\,\omega_m[k], \qquad \theta_r[k] = N_r\,\theta_m[k]. \tag{33}$$

The mechanical angle θ_m is supplied to the `Te` subsystem for the torque calculation of Equation (28) and is exported through output port 5 of the motor model for the encoder emulation. The electrical angle θ_r is supplied to the `StepperElectrical` subsystem, where it sets the phase of the back-EMF terms in Equations (15) and (16), and is exported through output port 3. The angle is not wrapped to a single revolution. Accumulating θ_m without bound preserves the multi-turn rotor position required by the encoder emulation,

and wrapping is unnecessary for the torque and back-EMF calculations because the sine and cosine functions are periodic.

Together, the `StepperMechanical` and `rotor angle` subsystems form a cascade of two discrete integrators from net torque to rotor position. Combined with the two phase-current states of the electrical model, the complete stepper motor model is described by the state vector $[\, i_a,\ i_b,\ \omega_m,\ \theta_m \,]^{\mathsf{T}}$. The signals of the mechanical model are summarized in Table 6, and its parameters in Table 7.

Table 6: Signals of the mechanical model.

Subsystem	Signal	Direction	Description
`Te`	i_a, i_b	Input	Phase currents (A)
	θ_m	Input	Mechanical rotor angle (rad)
	T_e	Output	Electromagnetic torque (N m)
`StepperMechanical`	T_e	Input	Electromagnetic torque (N m)
	T_L	Input	Load torque, motor input port 3 (N m)
	ω_m	Output	Rotor speed, motor output port 4 ($\mathrm{rad\,s^{-1}}$)
`rotor angle`	ω_m	Input	Rotor speed ($\mathrm{rad\,s^{-1}}$)
	θ_m	Output	Mechanical angle, motor output port 5 (rad)
	θ_r	Output	Electrical angle, motor output port 3 (rad)

Table 7: Parameters of the mechanical model.

Symbol	Simulink Block	Description
K_m	`Te`	Torque constant, $0.377\,\mathrm{N\,m\,A^{-1}}$
N_r	`RT` gain	Number of rotor teeth, 50
J	`1/J` gain	Total rotor and load inertia ($\mathrm{kg\,m^2}$)
B_m	`Bm` gain	Viscous damping coefficient ($\mathrm{N\,m\,s\,rad^{-1}}$)
T_s	Discrete-Time Integrators	Simulation sample time (s)

PCB Model

The PCB model emulates the measurement interface between the stepper motor and the FPGA. The stepper motor model produces the rotor angle and phase currents as double-precision signals, while the physical controller receives these quantities only through the sensors and converters on the PCB: a quadrature encoder for rotor position, and shunt amplifiers sampled by analog-to-digital converters (ADCs) for the phase currents. The PCB model converts the motor outputs into the same digital signals, with the same data types, resolution, and offsets, that the FPGA receives from the board. This allows the decoding logic, fixed-point data types, and measurement quantization of the controller to be exercised in simulation with the same form of inputs it will process in hardware.

Encoder Emulation

Rotor position is measured on the physical system with a 5000-line incremental quadrature encoder mounted to the motor shaft. The encoder produces two square-wave channels, A and B, displaced from one another by a quarter cycle, and an index channel, Z, that pulses once per revolution. Decoding every edge of both channels yields four counts per line, so one mechanical revolution corresponds to

$$N_c = 4N_{\text{line}} = 4 \times 5000 = 20000 \tag{34}$$

counts. The `Convert Mechanical Angle to Encoder Pulses` subsystem, shown in Figure , receives the mechanical rotor angle θ_m from the `rotor angle` subsystem of the stepper motor model and generates the three Boolean encoder channels through four operations.

- **Angle scaling.** A gain block converts the mechanical angle from radians to encoder

counts by multiplying by $N_c/2\pi$.

- **Integer conversion.** A Data Type Conversion block casts the scaled angle from double precision to a signed 32-bit integer, quantizing the continuous angle to a whole number of encoder counts. This count is the position an ideal encoder interface would accumulate, and all three output channels are derived from it.

- **Quadrature generation.** Bitwise AND blocks with the masks `0x1` and `0x2` extract the two least significant bits of the count. Channel B is taken directly from the second bit, and channel A is formed as the exclusive OR of the first and second bits.

- **Index generation.** A Math Function block computes the count modulo N_c, and a comparison block asserts the index channel when the result is zero.

Combining these operations, the encoder count at sample k is

$$n[k] = \mathcal{Q}\left(\frac{N_c}{2\pi}\,\theta_m[k]\right), \tag{35}$$

where $\mathcal{Q}(\cdot)$ denotes the integer conversion of the Data Type Conversion block. Writing $b_0[k]$ and $b_1[k]$ for the first and second bits of $n[k]$, the encoder channels are

$$B[k] = b_1[k], \qquad A[k] = b_0[k] \oplus b_1[k], \qquad Z[k] = \begin{cases} 1, & n[k] \bmod N_c = 0, \\ 0, & \text{otherwise.} \end{cases} \tag{36}$$

Equation (36) converts the two low bits of the binary count into the two-bit Gray code produced by a quadrature encoder, summarized in Table 8. As the count increases, the channel state advances through the sequence $(A, B) = (0,0) \to (1,0) \to (1,1) \to (0,1)$, so

that channel A leads channel B by a quarter cycle. When the count decreases, the sequence reverses and channel B leads. Exactly one channel changes on each count, which is the property the decoder relies on to determine the direction of rotation. Because the count is stored in two's complement form, its two low bits continue the same four-state cycle as the count passes through zero and becomes negative. A reversal of direction across the zero position therefore produces a continuous quadrature sequence with no spurious edges.

Table 8: Quadrature states generated from the encoder count.

$n \bmod 4$	b_1	b_0	A	B
0	0	0	0	0
1	0	1	1	0
2	1	0	1	1
3	1	1	0	1

The index channel is asserted for a single count whenever the rotor is at the reference zero position or a whole number of revolutions from it, which matches the once-per-revolution index pulse of the physical encoder. Because the mechanical angle is not wrapped to a single revolution, the count accumulates the multi-turn position of the rotor. The signed 32-bit count can represent approximately $2^{31}/N_c \approx 107{,}000$ revolutions before overflow, which is far beyond the travel of any simulation.

The resolution of the encoder is one count, or

$$\Delta\theta_m = \frac{2\pi}{N_c} = 3.14 \times 10^{-4}\,\text{rad} \quad (0.018^\circ). \tag{37}$$

The controller recovers the mechanical angle as $\hat{\theta}_m = 2\pi n/N_c$, and the electrical angle used in the Park transform of Equation (19) as $\hat{\theta}_e = N_r\hat{\theta}_m$, following Equation (18). One electrical revolution therefore spans $N_c/N_r = 400$ counts, and the electrical angle resolution is 0.9°. An error of δ in the electrical angle rotates the measured current vector by δ. When

the controller regulates the measured d-axis current to zero, the actual d-axis current is approximately $\sin\delta \approx 1.6\%$ of the current magnitude for a one-count error, and the torque-producing current is reduced by the factor $\cos\delta > 0.9998$. Both effects are negligible.

The emulated encoder also places an upper limit on the rotor speed that can be represented without loss of counts. From Equation (33), the mechanical angle advances by $T_s\,\omega_m$ each sample, so the count advances by approximately $N_c T_s \omega_m / 2\pi$ counts per sample. If the count advances by two or more in a single sample, both channels change together and the direction of motion cannot be determined from the channel states. The emulated channels therefore remain valid only while

$$|\omega_m| < \frac{2\pi}{N_c T_s} = \frac{2\pi}{20000 \times 20\,\mu\text{s}} \approx 15.7\,\text{rad}\,\text{s}^{-1} \quad (150\,\text{rpm}). \tag{38}$$

This limit is an artifact of updating the rotor angle in discrete steps. The physical encoder produces continuous edges, and the FPGA samples them at its system clock, which is many times faster than the maximum count rate of the motor.

Table 9: Signals of the encoder emulation.

Signal	Type	Direction	Description
θ_m	double	Input	Mechanical rotor angle (rad)
n	int32	Internal	Accumulated encoder count
A (`enc_a`)	Boolean	Output	Quadrature channel A
B (`enc_b`)	Boolean	Output	Quadrature channel B
Z (`enc_z`)	Boolean	Output	Index, once per revolution

Current Sensing

The phase currents are measured on the physical system with a shunt resistor in series with each winding. The voltage across the shunt is amplified by an INA241 current-sense amplifier and digitized by a 16-bit ADC. Because the winding current reverses direction, the

Table 10: Parameters of the encoder emulation.

Symbol	MATLAB Variable	Description
N_{line}	—	Encoder lines per revolution, 5000
N_c	`CountsPerRev`	Counts per mechanical revolution, 20000
N_c/N_r	`electricalCountsPerRev`	Counts per electrical revolution, 400
$\Delta\theta_m$	`mechanicalRadiansPerCount`	Mechanical angle per count, 3.14×10^{-4} rad

amplifier output is offset to half of the ADC reference voltage, so that both current polarities fall within the input range of the ADC. One `ADC Emulator` subsystem, shown in Figure , is used for each phase. It receives the phase current i_x from the stepper motor model and produces the 16-bit ADC reading through four operations.

- **Shunt and amplifier gain.** A gain block multiplies the phase current by the product of the shunt resistance and the amplifier gain, $K_{cs} = R_{sh}\, G_{cs}$, converting current into the amplifier output voltage. The shunt and amplifier are modeled as an ideal gain with no bandwidth limit, gain error, or offset error.

- **Amplifier offset.** A Sum block adds the constant offset $V_{\text{off}} = V_{\text{ref}}/2$, which models the reference voltage applied to the amplifier according to the INA241 datasheet.

- **Voltage to ADC scaling.** A gain block converts the amplifier output voltage into ADC counts by multiplying by $K_{\text{adc}} = 2^N/V_{\text{ref}}$, where $N = 16$ is the ADC resolution in bits.

- **Quantization.** A Data Type Conversion block casts the result from double precision to the unsigned 16-bit type `fixdt(0,16,0)`, producing the integer reading the ADC reports to the FPGA.

Combining these operations, the amplifier output voltage and ADC reading for phase x at sample k are

$$v_{cs,x}[k] = K_{cs}\, i_x[k] + V_{\text{off}}, \qquad N_x[k] = \mathcal{Q}_{16}(K_{\text{adc}}\, v_{cs,x}[k])\,, \tag{39}$$

where $\mathcal{Q}_{16}(\cdot)$ denotes the conversion to an unsigned 16-bit integer. Substituting $V_{\text{off}} = V_{\text{ref}}/2$ gives

$$N_x[k] = \mathcal{Q}_{16}\big(K_{\text{adc}} K_{cs}\, i_x[k] + 2^{N-1}\big)\,, \tag{40}$$

so that zero current maps to the mid-scale reading $2^{N-1} = 32768$, positive current maps above mid-scale, and negative current maps below it. The input range of the ADC, $0 \le v_{cs,x} < V_{\text{ref}}$, sets the measurable current range and the current resolution as

$$|i_x| < \frac{V_{\text{ref}}}{2K_{cs}} = 2.5\,\text{A}, \qquad \Delta i = \frac{V_{\text{ref}}}{2^N K_{cs}} = 76.3\,\mu\text{A}. \tag{41}$$

Currents outside this range drive the amplifier output beyond the input range of the ADC, and the reading clips at 0 or $2^N - 1$. The resolution is far finer than the PWM current ripple of approximately 89 mA from Equation (14), which spans roughly 1170 ADC counts. The ADC therefore resolves the ripple itself, and the measured current depends on the point in the PWM period at which the sample is taken. The quantization error of at most one count is negligible by comparison.

The controller recovers the phase current by inverting Equation (40),

$$\hat{\imath}_x[k] = \frac{N_x[k] - 2^{N-1}}{K_{\mathrm{adc}} K_{cs}}. \tag{42}$$

Because the mid-scale offset 2^{N-1} is a power of two, it is removed in the FPGA with a subtraction of an unsigned integer constant, and because $K_{cs} = 1\,\mathrm{V\,A^{-1}}$, the remaining conversion is a single multiplication by $1/K_{\mathrm{adc}}$.

The amplifier and ADC are modeled as ideal. Amplifier bandwidth, common-mode transients at the PWM switching edges, the input filters, and ADC noise, nonlinearity, and conversion latency are omitted. Unlike the motor-side disturbances, errors in the current measurement are not rejected by the Active Disturbance Rejection Control system. The current loop regulates the measured current, so a gain or offset error in the measurement becomes an error in the actual phase current. For this reason, the shunt resistance and amplifier offset used in the final system should be calibrated values measured on the PCB rather than nominal values.

Table 11: Signals of the current sensing emulation.

Signal	Type	Direction	Description
i_a, i_b	double	Input	Phase currents from the stepper motor model (A)
$v_{cs,x}$	double	Internal	Amplifier output voltage (V)
N_a, N_b	uint16	Output	ADC readings, PCB model output ports 1 and 2

Load Cell Emulation

Control System

The control system regulates the position of the stepper motor through field-oriented control, in which the phase currents are measured, transformed into the rotor-fixed d–q frame,

Table 12: Parameters of the current sensing emulation.

Symbol	MATLAB Variable	Description
R_{sh}	`Shunt_Resistance`	Shunt resistance, $0.1\,\Omega$
G_{cs}	`Shunt_Amp_Gain`	Amplifier gain, $10\,\mathrm{V\,V^{-1}}$
K_{cs}	`ADC_Emulated_conversion_factor`	Current-to-voltage gain, $1\,\mathrm{V\,A^{-1}}$
V_{off}	`Shunt_Amp_Offset`	Amplifier output offset, 2.5 V
V_{ref}	`ADC_VREF`	ADC reference voltage, 5 V
N	`ADC_Precision`	ADC resolution, 16 bits
K_{adc}	`VoltagetoADCValue`	Voltage-to-count gain, 13 107.2 counts/V

and regulated so that the motor produces torque only through the quadrature current. The controller is implemented in Simulink in fixed-point arithmetic so that it can be generated as HDL code for the FPGA. Every input it receives is in the form produced by the PCB model of Section 2: unsigned 16-bit ADC readings for the phase currents and Boolean quadrature channels from the encoder. Every output is a Boolean PWM and direction signal for each H-Bridge, as described in Section 2.

Control Architecture

The motor is controlled by a cascade of three loops, shown in Figure . The position loop, a proportional controller with gain $K_{P\theta}$, produces a velocity reference. The velocity loop, a first-order Active Disturbance Rejection Controller (ADRC), computes the quadrature current command i_q^*. Two current loops, also first-order ADRCs, compute the direct and quadrature voltage commands v_d^* and v_q^*. The signal path through the controller consists of five stages.

- **Feedback processing.** The ADC readings are decoded into phase currents in amperes, and the encoder channels are decoded into the mechanical position, the electrical angle, and the rotor velocity.

- **Position loop.** The position error is multiplied by $K_{P\theta}$ to form the velocity reference ω_m^*.
- **Velocity loop.** The velocity ADRC computes the quadrature current command i_q^* from ω_m^* and the measured velocity.
- **Current loops.** The measured phase currents are transformed into i_d and i_q with the Park transform, and two current ADRCs compute v_d^* and v_q^* that drive i_d to zero and i_q to i_q^*.
- **Voltage output.** The inverse Park transform converts v_d^* and v_q^* into the phase voltage commands, which the PWM generators convert into the gate and direction signals of each H-Bridge.

All three loops are executed at the controller sample time $T_s = 20\,\mu s$, which equals the PWM period, so that each loop is updated once per PWM cycle. The closed-loop and observer bandwidths are summarized in Table 13. Each loop is roughly an order of magnitude slower than the loop it commands: the current loop is 13.3 times faster than the velocity loop, and the velocity loop is 5 times faster than the position loop. This separation allows each inner loop to be treated as an ideal actuator by the loop around it, with the residual inner-loop dynamics absorbed by the observer of the outer ADRC.

Table 13: Closed-loop bandwidth ω_{CL} and observer bandwidth ω_{ESO} of the control loops.

Loop	Controller	ω_{CL}	ω_{ESO}
Position	Proportional	$150\,\mathrm{rad\,s^{-1}}$	—
Velocity	First-order ADRC	$750\,\mathrm{rad\,s^{-1}}$	$6000\,\mathrm{rad\,s^{-1}}$
Current (d and q)	First-order ADRC	$10\,000\,\mathrm{rad\,s^{-1}}$	$15\,000\,\mathrm{rad\,s^{-1}}$

The controller gains and ADRC coefficients are supplied to the controller subsystem as inputs rather than as internal constants. In the FPGA implementation these inputs can

be held in registers written at run time, so that the controller can be re-tuned without regenerating or re-synthesizing the HDL.

Feedback Signal Processing

Current Decoding The `adc_phase_a_decoder` and `adc_phase_b_decoder` subsystems invert the current sensing chain of Section 2. Each decoder receives the unsigned 16-bit ADC reading N_x, passes it through a Rate Transition block onto the controller rate, converts it to a signed type, subtracts the mid-scale offset $N_{\text{off}} = 2^{15} = 32768$, and scales the result to amperes:

$$\hat{\imath}_x[k] = K_{\text{dec}}\Big(N_x[k] - N_{\text{off}}\Big), \qquad K_{\text{dec}} = \frac{1}{K_{\text{adc}} K_{cs}} = 7.63 \times 10^{-5}\,\text{A}, \tag{43}$$

which is Equation (42) with K_{dec} expressed as amperes per count. Because the offset is a power of two, the subtraction is a single integer operation. The implemented gain of 1/13107 differs from the exact value 1/13107.2 by a relative error of 1.5×10^{-5}, which amounts to 38 µA at the full-scale current of 2.5 A, less than half of one ADC count.

Encoder Decoding The `Encoder Decoder` subsystem, shown in Figure , reconstructs the rotor position from the quadrature channels A and B of Equation (36). It is the inverse of the encoder emulation of Section 2 and is built entirely from Boolean logic, comparisons, and integer additions so that it maps directly to FPGA logic. The decoder operates in five stages.

- **Edge detection.** Each channel is compared with its value at the previous sample through a unit delay and an exclusive OR, producing the edge flags $e_A[k] = A[k] \oplus A[k-1]$ and $e_B[k] = B[k] \oplus B[k-1]$.

- **Direction detection.** The direction of rotation is determined by comparing the current state of channel A with the previous state of channel B, $d_f[k] = A[k] \oplus B[k-1]$. For every valid transition in Table 8, $d_f = 1$ when the count is increasing and $d_f = 0$ when it is decreasing.

- **Count detection.** A valid count occurs when exactly one channel changes, $t[k] = e_A[k] \oplus e_B[k]$. When both channels change in the same sample, $e_A[k] \wedge e_B[k]$ flags an illegal transition, which corresponds to the encoder speed limit of Equation (38) being exceeded.

- **Position accumulation.** On each valid count, the position count is incremented when $d_f = 1$ and decremented otherwise. A preset input overwrites the count with a supplied value.

- **Electrical count.** A second counter tracks the position within one electrical revolution. It increments or decrements on each valid count in the same way, but rolls over from 399 to 0 when increasing and from 0 to 399 when decreasing. A zeroing input resets it to zero.

Combining these operations, the position count $\hat{n}$ and electrical count $\hat{n}_e$ are updated as

$$\hat{n}[k+1] = \begin{cases} n_{\text{preset}}, & \text{preset enabled}, \\ \hat{n}[k] + t[k]\left(2d_f[k] - 1\right), & \text{otherwise}, \end{cases} \tag{44}$$

$$\hat{n}_e[k+1] = \begin{cases} 0, & \text{zeroing enabled}, \\ \left(\hat{n}_e[k] + t[k]\left(2d_f[k] - 1\right)\right) \bmod \dfrac{N_c}{N_r}, & \text{otherwise}, \end{cases} \tag{45}$$

where the modulo operation is implemented by the comparison and rollover logic rather than by a division. The mechanical and electrical angles used by the controller are then

$$\hat{\theta}_m[k] = \frac{2\pi}{N_c}\,\hat{n}[k], \qquad \hat{\theta}_e[k] = \frac{2\pi}{N_c/N_r}\,\hat{n}_e[k], \tag{46}$$

with $N_c = 20000$ counts per mechanical revolution from Equation (34) and $N_c/N_r = 400$ counts per electrical revolution. Tracking the electrical count separately keeps the electrical angle within a single revolution without a modulo operation on the full position count, and is consistent with Equation (18) because $N_r\hat{\theta}_m$ and $\hat{\theta}_e$ differ only by whole electrical revolutions when both counters are zeroed together. The radians-per-count constants are stored with 64 fractional bits so that the conversion adds no error beyond the encoder resolution.

The electrical angle is meaningful only if $\hat{n}_e = 0$ coincides with the alignment of the rotor d-axis with phase A. On the physical system the rotor is first aligned by energizing phase A, after which the zeroing input sets $\hat{n}_e = 0$.

Velocity Estimation The velocity estimate is smoothed by a first-order low-pass filter before it is used by the velocity ADRC,

$$\hat{\omega}_f[k] = \hat{\omega}_f[k-1] + a_f\Big(\hat{\omega}_m[k] - \hat{\omega}_f[k-1]\Big), \qquad a_f = 1 - e^{-\omega_f T_s}, \tag{47}$$

which is the exact discretization of the continuous filter $\omega_f/(s+\omega_f)$. With $\omega_f = 5000\,\mathrm{rad\,s^{-1}}$, the filter coefficient is $a_f = 0.0952$ and the filter time constant is 0.2 ms, which is short compared with the velocity loop time constant $1/\omega_{CL} = 1.33\,\mathrm{ms}$. The filter bandwidth is below the velocity observer bandwidth of $6000\,\mathrm{rad\,s^{-1}}$, so the filter lag lies within the

bandwidth that the observer tracks and appears to it as unmodeled plant dynamics.

Rotor-Frame Currents The decoded phase currents are transformed into the rotor frame by the Park transform of Equation (19), evaluated with the measured electrical angle,

$$\hat{\imath}_d = \hat{\imath}_a \cos\hat{\theta}_e + \hat{\imath}_b \sin\hat{\theta}_e, \qquad \hat{\imath}_q = -\hat{\imath}_a \sin\hat{\theta}_e + \hat{\imath}_b \cos\hat{\theta}_e. \tag{48}$$

Because the motor has two phases displaced by 90°, the phase currents are already orthogonal components, and no Clarke transform is required before the Park rotation.

Plant Models for Control Design

Each ADRC is designed around a first-order model of the form

$$\dot{y} = b_0\, u + f, \tag{49}$$

where y is the controlled output, u is the control input, b_0 is the input gain, and f is the total disturbance [**? ?**]. The total disturbance collects every term in the true dynamics other than $b_0 u$, including external loads, coupling, parameter error, and unmodeled dynamics, and is estimated online by an extended state observer. Only b_0 must be known in advance.

Velocity Loop Input Gain Setting $i_d = 0$ and substituting Equation (25) into Equation (22) gives the velocity dynamics with the quadrature current as the input:

$$\dot{\omega}_m = \frac{K_m}{J}\, i_q - \frac{B_m\, \omega_m + T_L}{J}. \tag{50}$$

Comparing with Equation (49), with $y = \omega_m$ and $u = i_q^*$, the input gain is the known part of the plant,

$$b_0 = \frac{N_r\,\psi_m}{J} = \frac{K_m}{J} = \frac{K_m}{J_m + J_s + J_n}. \tag{51}$$

Using $K_m = 0.3771\,\mathrm{N\,m\,A^{-1}}$ and $J = 7.002 \times 10^{-5}\,\mathrm{kg\,m^2}$,

$$b_0 = \frac{0.3771\,\mathrm{N\,m\,A^{-1}}}{7.002 \times 10^{-5}\,\mathrm{kg\,m^2}} = 5385\,\mathrm{rad\,s^{-2}\,A^{-1}}. \tag{52}$$

The units show the role of b_0: it converts a current command in A into an angular acceleration in $\mathrm{rad\,s^{-2}}$. The damping and load terms are not included in b_0. They are absorbed into the total disturbance

$$f_\omega = -\frac{B_m\,\omega_m + T_L}{J} + \frac{K_m}{J}\left(i_q - i_q^*\right) + \Delta_\omega, \tag{53}$$

where the second term is the tracking error of the current loop and Δ_ω collects unmodeled effects such as friction, detent torque, and the ball screw load. This keeps b_0 dependent only on the torque constant and the inertia, which are the best known parameters of the system.

Current Loop Input Gain Rearranging Equations (20) and (21) for the current derivatives gives

$$\frac{di_d}{dt} = \frac{1}{L}\,v_d + \underbrace{\frac{-R\,i_d + N_r\,\omega_m L\,i_q}{L}}_{f_d}, \tag{54}$$

$$\frac{di_q}{dt} = \frac{1}{L}\,v_q + \underbrace{\frac{-R\,i_q - N_r\,\omega_m L\,i_d - N_r\,\omega_m\,\psi_m}{L}}_{f_q}. \tag{55}$$

Both axes therefore have the form of Equation (49) with the same input gain,

$$b_i = \frac{1}{L} = \frac{1}{2.7\,\mathrm{mH}} = 370.4\,\mathrm{A\,s^{-1}\,V^{-1}}. \tag{56}$$

The winding resistance, the speed-dependent cross-coupling between the axes, and the back EMF are all contained in the total disturbances f_d and f_q. The current loops therefore require no explicit decoupling or back-EMF feedforward terms, which are normally needed in field-oriented control and which would require accurate values of R and ψ_m.

Discrete-Time First-Order ADRC

Both the velocity loop and the current loops use the same discrete-time first-order ADRC, implemented in the minimum-footprint form of Herbst and Madonski [**?**]. This section derives the controller from the plant model of Equation (49) and shows how the minimum-footprint form is obtained from it.

Extended State Observer and Control Law Treating the total disturbance as an additional state, $\mathbf{x} = [\,y,\ f\,]^{\mathsf{T}}$, and discretizing Equation (49) with the forward Euler method at the sample time T_s gives

$$\mathbf{x}[k+1] = \underbrace{\begin{bmatrix} 1 & T_s \\ 0 & 1 \end{bmatrix}}_{\mathbf{\Phi}} \mathbf{x}[k] + \underbrace{\begin{bmatrix} b_0 T_s \\ 0 \end{bmatrix}}_{\mathbf{\Gamma}} u[k], \qquad y[k] = \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{\mathbf{c}^\mathsf{T}} \mathbf{x}[k], \tag{57}$$

in which the disturbance is modeled as constant over one sample. The extended state observer is implemented as a current estimator, which corrects the prediction from the previous sample with the measurement taken at the present sample:

$$\hat{\mathbf{x}}[k] = \big(\mathbf{I} - \mathbf{l}\,\mathbf{c}^\mathsf{T}\big)\Big(\mathbf{\Phi}\,\hat{\mathbf{x}}[k-1] + \mathbf{\Gamma}\,u_{\mathrm{lim}}[k-1]\Big) + \mathbf{l}\,y[k], \qquad \mathbf{l} = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}. \tag{58}$$

The observer is driven by the limited control signal u_{lim}, which is the control signal actually applied to the plant after saturation, rather than the unlimited controller output. Placing both observer poles at $z_{ESO} = e^{-\omega_{ESO} T_s}$ requires

$$l_1 = 1 - z_{ESO}^2, \qquad l_2 = \frac{\big(1 - z_{ESO}\big)^2}{T_s}. \tag{59}$$

The control law cancels the estimated disturbance and applies proportional feedback on the estimated output,

$$u[k] = \frac{u_0[k] - \hat{f}[k]}{b_0}, \qquad u_0[k] = k_1\big(r[k] - \hat{y}[k]\big), \tag{60}$$

where r is the reference. When the estimates converge, substituting Equation (60) into Equation (57) reduces the plant to $y[k+1] = y[k] + T_s k_1 (r[k] - y[k])$, a first-order system with its pole at $z = 1 - k_1 T_s$. Placing this pole at $z_{CL} = e^{-\omega_{CL} T_s}$ gives

$$k_1 = \frac{1 - z_{CL}}{T_s}. \tag{61}$$

The design is therefore fully specified by four quantities: the input gain b_0, the closed-loop bandwidth ω_{CL}, the observer bandwidth $\omega_{ESO} = k_{ESO}\,\omega_{CL}$, and the sample time T_s. This is the bandwidth parameterization of Gao [**?**], applied to the discrete poles [**?**].

Minimum-Footprint Implementation Eliminating the observer states from Equations (58) and (60) gives an equivalent controller in transfer-function form, which requires only two delay elements and seven multiplications:

$$U(z) = \frac{k_1}{b_0}\,R(z) - C_y(z)\,Y(z) + C_u(z)\,U_{\mathrm{lim}}(z), \tag{62}$$

$$C_y(z) = \frac{\beta_0 + \beta_1 z^{-1}}{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}, \qquad C_u(z) = \frac{\gamma_0 z^{-1} + \gamma_1 z^{-2}}{1 + \alpha_1 z^{-1} + \alpha_2 z^{-2}}, \tag{63}$$

with the coefficients

$$\begin{aligned}
&\alpha_1 = -2z_{ESO}, \qquad \alpha_2 = z_{ESO}^2, \\
&\beta_0 = \frac{z_{CL} z_{ESO}^2 - 2z_{ESO} - z_{CL} + 2}{T_s b_0}, \\
&\beta_1 = \frac{2z_{CL} z_{ESO} - 2z_{CL} z_{ESO}^2 + z_{ESO}^2 - 1}{T_s b_0}, \\
&\gamma_0 = z_{CL} z_{ESO}^2 - 2z_{ESO} + 1, \qquad \gamma_1 = z_{ESO}^2\big(1 - z_{CL}\big),
\end{aligned} \tag{64}$$

The coefficients are computed offline by the function `MFP_ADRC_1st_params_multirate`. Substituting Equations (59) and (61) into Equations (58) and (60) and simplifying reproduces

Equations (62) through (64) exactly, which confirms that the minimum-footprint controller and the observer-based controller are the same controller. The denominator $1 + \alpha_1 z^{-1} + \alpha_2 z^{-2} = (1 - z_{ESO} z^{-1})^2$ contains the two observer poles. The coefficients β_0 and β_1 carry the units of u per unit of y, while α_i and γ_i are dimensionless.

The `MFP_ADRC_1st` subsystems, shown in Figure , implement Equation (62) as the difference equations

$$\begin{aligned} w[k] &= x_1[k] - \beta_0\, y[k], \\ u[k] &= \frac{k_1}{b_0}\, r[k] + w[k], \\ x_1[k+1] &= x_2[k] - \beta_1\, y[k] + \gamma_0\, u_{\mathrm{lim}}[k] - \alpha_1\, w[k], \\ x_2[k+1] &= \gamma_1\, u_{\mathrm{lim}}[k] - \alpha_2\, w[k], \end{aligned} \tag{65}$$

where x_1 and x_2 are the outputs of the two unit delays and w is the feedback portion of the control signal. The reference feedforward term $k_1 r / b_0$ and the feedback term w are summed and converted to the output data type. Because $y[k]$ enters $u[k]$ directly through β_0, the controller responds to a new measurement in the same sample, which is the advantage of the current estimator over a predictive one. Because $u_{\mathrm{lim}}[k]$ affects only the delay states, feeding the limited signal back to the controller does not create an algebraic loop.

Closed-Loop Properties For the nominal plant of Equation (57) without saturation, the closed-loop transfer functions from the reference and from the total disturbance to the output are

$$\frac{Y(z)}{R(z)} = \frac{1 - z_{CL}}{z - z_{CL}}, \qquad \frac{Y(z)}{F(z)} = \frac{T_s\,(z-1)\left(z - z_{CL} z_{ESO}^2\right)}{\left(z - z_{CL}\right)\left(z - z_{ESO}\right)^2}. \tag{66}$$

The reference response is exactly first order with bandwidth ω_{CL}, and the observer poles

cancel from it entirely, so the observer bandwidth can be chosen for disturbance rejection without affecting the reference response. The disturbance response has a zero at $z = 1$, so a constant disturbance, such as a constant load torque or a constant back EMF, is rejected completely in steady state. The characteristic polynomial, $(z - z_{CL})(z - z_{ESO})^2$, shows that the controller and observer poles can be placed independently.

The integral action responsible for the zero at $z = 1$ resides in the feedback of u_{lim} through $C_u(z)$. Solving Equation (62) for U gives a factor of $1/(1 - C_u(z))$, which has a pole at $z = 1$ only when $C_u(1) = 1$, or equivalently

$$\gamma_0 + \gamma_1 = 1 + \alpha_1 + \alpha_2 = \big(1 - z_{ESO}\big)^2. \tag{67}$$

The coefficients of Equation (64) satisfy this identity exactly. When the control signal saturates, u_{lim} stops following u, the loop through $C_u(z)$ no longer integrates, and the controller is protected from integrator windup without any additional anti-windup logic [**?**].

Because the observer absorbs modeling error into f, b_0 need not be exact. If the true input gain differs from b_0, the term $(b - b_0)\, u$ is estimated as part of the total disturbance. A value of b_0 larger than the true value reduces the effective loop gain, and a smaller value increases it, so b_0 is a robustness and tuning parameter rather than a quantity that must be identified precisely.

Position Loop

The `Position` subsystem implements a proportional controller on the mechanical position,

$$\omega_m^*[k] = \text{sat}_{\omega_{\max}}\Big(K_{P\theta}\big(\theta_m^*[k] - \hat{\theta}_m[k]\big)\Big), \tag{68}$$

where θ_m^* is the position target and the velocity reference is limited to $\pm\omega_{\max}$. Because the velocity loop is much faster than the position loop, it can be approximated as a unity gain, and the position loop reduces to $\dot{\theta}_m = K_{P\theta}(\theta_m^* - \theta_m)$, a first-order response with bandwidth $K_{P\theta} = 150\,\mathrm{rad\,s^{-1}}$. A proportional controller is sufficient for the position loop because the velocity loop provides the integrating plant: the position is the integral of the regulated velocity, so a constant position error cannot persist. Load torque on the axis is rejected by the velocity ADRC before it can produce a steady-state position error.

Velocity Loop

The velocity loop applies the controller of Section 2 with $y = \hat{\omega}_f$, $r = \omega_m^*$, and $u = i_q^*$. The input gain is b_0 from Equation (52), the closed-loop bandwidth is $\omega_{CL} = 750\,\mathrm{rad\,s^{-1}}$, and the observer bandwidth is $k_{ESO} = 8$ times the closed-loop bandwidth, or $6000\,\mathrm{rad\,s^{-1}}$. The resulting coefficients are listed in Table 14.

The output of the velocity ADRC is limited to $\pm I_{\max}$ before it is used as the current command, and the limited command is fed back to the controller as u_{lim}. The limit protects the motor windings and keeps the phase currents within the $\pm 2.5\,\mathrm{A}$ measurement range of Equation (41).

Current Loops

The `current error` subsystem contains two instances of the controller of Section 2, one for each axis. The direct-axis loop uses $y = \hat{\imath}_d$, $r = i_d^* = 0$, and $u = v_d^*$. The quadrature-axis loop uses $y = \hat{\imath}_q$, $r = i_q^*$, and $u = v_q^*$. Both loops use the input gain b_i of Equation (56), a closed-loop bandwidth of $\omega_{CL} = 10\,000\,\mathrm{rad\,s^{-1}}$, and an observer bandwidth of $k_{ESO} = 1.5$ times the closed-loop bandwidth, or $15\,000\,\mathrm{rad\,s^{-1}}$. Because the two axes have identical models, they share one set of coefficients, listed in Table 14. Each controller output is

saturated before leaving the subsystem, and the voltage actually applied on each axis is returned to the controller as u_{lim}, as described in Section 2.

At the current-loop bandwidth, $\omega_{CL}T_s = 0.2$ and $\omega_{ESO}T_s = 0.3$, so the discrete poles $z_{CL} = 0.819$ and $z_{ESO} = 0.741$ remain well inside the unit circle and away from the origin. The 20 µs sample time is therefore fast enough for the chosen bandwidths without the discretization dominating the response.

Table 14: ADRC design parameters and coefficients, computed with $T_s = 20\,\mu\mathrm{s}$.

Quantity	Velocity loop	Current loops
b_0	$5385.5\,\mathrm{rad\,s^{-2}\,A^{-1}}$	$370.37\,\mathrm{A\,s^{-1}\,V^{-1}}$
ω_{CL}	$750\,\mathrm{rad\,s^{-1}}$	$10\,000\,\mathrm{rad\,s^{-1}}$
k_{ESO}	8	1.5
ω_{ESO}	$6000\,\mathrm{rad\,s^{-1}}$	$15\,000\,\mathrm{rad\,s^{-1}}$
z_{CL}	0.985112	0.818731
z_{ESO}	0.886920	0.740818
k_1	$744.40\,\mathrm{s^{-1}}$	$9063.5\,\mathrm{s^{-1}}$
k_1/b_0	$0.138\,224\,\mathrm{A\,s\,rad^{-1}}$	$24.4713\,\mathrm{V\,A^{-1}}$
α_1	-1.773841	-1.481636
α_2	0.786628	0.548812
β_0	$0.148\,210\,\mathrm{A\,s\,rad^{-1}}$	$20.1098\,\mathrm{V\,A^{-1}}$
β_1	$-0.146\,443\,\mathrm{A\,s\,rad^{-1}}$	$-18.4660\,\mathrm{V\,A^{-1}}$
γ_0	$1.075\,62 \times 10^{-3}$	$-3.230\,75 \times 10^{-2}$
γ_1	$1.171\,14 \times 10^{-2}$	$9.948\,27 \times 10^{-2}$
l_1	0.213372	0.451188
l_2	$639.35\,\mathrm{s^{-1}}$	$3358.8\,\mathrm{s^{-1}}$

Voltage Output and PWM Generation

The voltage commands are returned to the stationary frame by the inverse of the Park transform of Equation (48),

$$v_a^* = v_d^* \cos\hat{\theta}_e - v_q^* \sin\hat{\theta}_e, \qquad v_b^* = v_d^* \sin\hat{\theta}_e + v_q^* \cos\hat{\theta}_e, \tag{69}$$

and each phase voltage command is passed to an instance of the `PWM_Generator` subsystem, shown in Figure . The generator converts a signed voltage command into the gate and direction signals of Equation (11) through four operations.

- **Saturation.** The voltage command is limited to the supply voltage, $\pm V_s$, with $V_s = 48\,\mathrm{V}$.
- **Direction.** The direction signal is set when the limited command is less than or equal to zero, $d_x = (v_x^* \leq 0)$, which selects the negative polarity of the H-Bridge.
- **Duty computation.** The magnitude of the limited command is scaled and converted to an integer on-time n_{pwm}, in counts of a counter that runs N_{pwm} counts per PWM period. A Rate Transition block passes this value from the controller rate to the counter rate.
- **Comparison.** A free-running counter is compared with n_{pwm}, and the gate signal is high while the counter does not exceed the on-time.

The resulting duty ratio is $D_x = n_{\mathrm{pwm}}/N_{\mathrm{pwm}}$, and by Equation (12) the average voltage applied to the phase over one PWM period is

$$\bar{v}_x = V_s \frac{n_{\mathrm{pwm}}}{N_{\mathrm{pwm}}} \sigma_x, \qquad \sigma_x = 1 - 2d_x, \tag{70}$$

which is quantized in steps of V_s/N_{pwm}.

To give the current observers the voltage actually applied rather than the commanded voltage, each axis command v_d^* and v_q^* is also passed through an instance of the PWM generator, and the `Convert PWM to Voltage` subsystem reconstructs the applied voltage from the resulting on-time and direction using Equation (70). These reconstructed voltages

are the u_{lim} inputs of the two current ADRCs. They include both the saturation and the PWM quantization, so the observers do not attribute either effect to the total disturbance. The generators apply the voltage limit to each axis independently, whereas the physical limit applies to each phase. The reconstructed voltages therefore match the applied voltages, to within the PWM quantization, whenever no phase saturates, and are approximate when a phase does.

Fixed-Point Implementation

The controller is implemented in fixed-point arithmetic so that it can be generated as HDL. The data types used are summarized in Table 15, where `fixdt(s,w,f)` denotes a word of w bits with f fractional bits, signed if $s = 1$. Each type was chosen to cover the range of the signals it holds with a resolution finer than the measurement that produces them. For example, `PosType` resolves 1.5×10^{-5} rad, which is 20 times finer than one encoder count, and covers $\pm 32\,768$ rad, or about 5200 revolutions.

Table 15: Fixed-point data types of the controller.

Name	Type	Range	Resolution	Used for
`ParamType`	`fixdt(1,32,20)`	± 2048	9.5×10^{-7}	Controller parameters
`PosType`	`fixdt(1,32,16)`	± 32768	1.5×10^{-5}	Position
`CountType`	`fixdt(1,32,0)`	$\pm 2^{31}$	1	Encoder counts, ADC values
`SpeedLoopType`	`fixdt(1,32,16)`	± 32768	1.5×10^{-5}	Velocity ADRC
`CurrLoopType`	`fixdt(1,32,20)`	± 2048	9.5×10^{-7}	Current ADRC
`bitwiseOperator`	`fixdt(0,32,0)`	0 to $2^{32} - 1$	1	Bitwise logic
`VelType`	`fixdt(1,128,64)`	$\pm 9.2 \times 10^{18}$	5.4×10^{-20}	Velocity, radians per count

Quantizing the ADRC coefficients moves the controller and observer poles slightly from their design values. With the velocity coefficients quantized to 16 fractional bits, the largest relative coefficient error is 0.7%, in γ_0. The observer poles split from the designed double

pole at $6000\,\mathrm{rad\,s^{-1}}$ to $5948\,\mathrm{rad\,s^{-1}}$ and $6053\,\mathrm{rad\,s^{-1}}$, and the controller pole is unchanged to four significant figures. With the current coefficients quantized to 20 fractional bits, every coefficient is within a relative error of 5×10^{-6} of its design value. In both cases the closed-loop behavior is indistinguishable from the design.

The integral condition of Equation (67) is more sensitive than the pole locations, because the zero at $z = 1$ exists only if it holds exactly. When each coefficient is rounded independently, the velocity coefficients still satisfy it exactly, but the current coefficients miss it by one least significant bit, giving $C_u(1) = 0.999986$. The resulting steady-state current error is approximately $2\,\mu\mathrm{A}$ for the back EMF at $10\,\mathrm{rad\,s^{-1}}$, which is negligible. The integral action can be made exact for any data type by computing γ_1 after quantization as $\gamma_1 = 1 + \alpha_1 + \alpha_2 - \gamma_0$, using the quantized values of α_1, α_2, and γ_0.

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CONCLUSION

LaTeX produces documents that look great, automatically handles references and citations, and easily incorporates figures and tables. This is not a guide to LaTeX but rather an introduction to the MSU style. If you want more information about LaTeX many introductory guides can be found online.

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APPENDIX

EXAMPLE CODE

This work used the following MATLAB code to produce basically nothing. But, I'll include it here for the sake of an example.

```
% MATLAB code to say 'hello world'
disp('Hello world')
```